

Candle Shop

URL: <http://classwork.engr.oregonstate.edu:60903>

Feedback :

Feedback (Round 1):

Haoran Guo

The Group 54 team has designed a thoughtful POS system for a small candle shop. The overview effectively tells a story about the shop owner's needs and explains why a database solution would help. The entities are well-chosen, and the design includes useful features like scent and size lookup tables that keep the catalog organized.

A few areas could be improved. First, **the entity names should follow the project style guide and use plurals**. Currently, **the naming is inconsistent, some entities are singular while others are plural**. Second, **the document would benefit from a more structured outline that clearly lists each entity with its attributes and relationships in a consistent format**. Right now, the relationships are woven into the narrative, which makes it harder to quickly reference during implementation. Third, the overview mentions that the shop makes around 50 sales a day, which is a helpful numerical fact, **adding 1–2 more specific facts would further clarify the scope**

The ERD is included and shows all entities with their relationship. The M:N relationship between sales and candles is properly handled through the saleItems intersection table, satisfying the requirement for at least one many to many relationship. The citations section is marked "None", which is appropriate since this appears to be entirely original work.

Overall, this is strong and practical design. With consistent naming, a cleaner outline format, and a couple more numerical facts, this draft will be in great shape for the next steps. Nice work, Team 54!

This review is entirely Haoran Guo original work.

Charles Miles II

Hi Zander and Abderrahmane,

I enjoyed reading your Candle shop POS system. I believe your overview describes what the owner of the candle shop would like to do and the POS will help. The main fact is the owner is getting older and would like to keep up with employee information and customer information. You use at least four entities and list attributes with correct datatypes. **I would recommend when using unique I to increment one**. The naming convention for entities uses plural

names and singular for attributes but **could possibly use capitalization for attributes second name like firstNAME**, just a suggestion. The 1:M relationships look correct and more than one M:M relationship was identified like Employees to Candles and Customers to Employees.

This feedback review is the original work by Charles Miles.

Laura Riley

Group 54's overview of their proposed problem is very well thought out. It is a very realistic situation of an aging candle shop owner, with a real need for support in handling his inventory, sales, customers, and employees. The overview also lists specific facts, listing an average of 50 sales a day, and giving the background behind the location of the store and its clientele—regular customers in a small town.

There are six proposed entities: Employees, Candles, Customers, Sales, Scents, Sizes, and SaleItems. Each one has clearly listed attributes, types, constraints, and relationships between entities. These entities seem, to me, to have correctly formulated 1:M relationships, and there is a M:N relationship between Sales and Candles, with SaleItems acting as the intersection table between the two.

All entity names are plural with attributes singular, and are consistent throughout the document. All attribute names use snake case for naming.

I think this was really well done!

This review is original and written entirely by me.

Charles Lally

Hello Zander and Abderrahmane,

I really like your draft! The setting is charming, and I wish I could visit the shop you're describing.

Does the overview describe what problem is to be solved?

Yes, the overview describes the problem to be solved. The overview notes that this project will be a point-of-sale system for a small candle shop. The business owner's pre-existing system is described, and the overview makes it clear why a new system is necessary.

One small note, and this is a total nitpick, but **you might want to switch the 3rd and 4th sentence of the overview. Right now, the 3rd sentence references "His current volume" without indicating who you are referring to. The 4th sentence introduces "The store owner." I think that those sentences could be swapped without loss of clarity.**

Does the overview list specific facts?

Yes, the overview lists some specific facts. The business owner expects to make around 50 sales a day, and the setting for the business is very clear. Group 54 also added enough information to gather the personal motivations of the business owner: he wants to enjoy his retirement, live a stress-free life, and connect with his customers. From the standpoint of a developer implementing a system that would meet the business owner's wishes, this is all extremely useful information: you could formulate the business owner's personal motivations as non-functional criteria to really test whether you've created a product that will suit his needs.

Are at least four entities described, and does each one represent a single idea to be stored as a list?

Yes, at least four entities are described. There will be Employees, Customers, Candles, Sales, Scents, and Sizes. Each entity represents a single idea, which could be stored as a list of attributes.

Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints, and describe relationships between entities?

The outline of entity details describes the purpose of each entity, lists each entity's attributes, datatypes, and constraints, and describes the relationships between entities. There will be Employees, with enough vital information to identify them by name, seniority, and ID number; Customers, with all of the basic and specific information a store would want to gather, like names and contact information; Candles, which will have three separate relationships; Sales, with sufficient information for a POS system, as well as enough attributes that Sales could later be fed to a business intelligence system to draw conclusions about sales patterns; Scents, with basic information about the scent of the candles for sale in the shop; and Sizes, which reflects the marketing names and actual volume of the various candles.

Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database?

Yes, the 1:M relationships are correctly formulated. One Employee can process many Sales, one Customer can initiate many Sales, one Scent can be included in many Candles, many Candles can be one Size, and many Candles can be sold in one Sale. There is at least one M:M relationship, between Sales and Candles, and Group 54 detailed how that relationship will work in the SalesItems transaction table. The ERD does present a logical view of the database.

Is there consistency in a) naming between overview and entity/attributes b) entities plural,

attributes singular c) use of capitalization for naming?

Yes, there is consistency in naming between the Overview and description of entities/attributes. Sales, customers, candles, sizes, scents, and employees are all introduced in the Overview, and the singular and plural spelling of each is consistent later. The use of capitalization is consistent throughout: there are no capitals in attribute names, and the capitalization of entities is consistent once they are formally identified as entities in a database.

Citations

This peer review is my work entirely. I did not consult outside sources or AI in writing this peer review.

Thank you - Charles

Actions based on feedback:

1. Tables were made plural
 - a. SaleItems → SalesItems
 - b. Sale → Sales (when referenced in text)
2. Keys converted from snake case to camel case
 - a. ie. customer_id → customerId
3. Auto increment was added to primary ids
4. Updated ER diagram with changes
5. Written representation of tables was cleaned up and descriptions separated.
6. Owner was introduced before being referenced
 - a. 3rd and 4th sentence of overview switched places.
7. Added additional numerical facts to the overview.
8. Reformatted complete document

Feedback (Round 2):

Mark Ferrara

Hello, Group 54! The overview was hilarious. Maybe all the candle fumes are not good for his aging brain!

Anyway, I did not find many errors in your PDF. Just a couple of things caught my attention. The table names are not capitalized in the ERD, but they are capitalized in the schema. Also, the ERD made me a little cross-eyed because of all the lines. However, I do like that you labeled the relationship lines. Maybe I should do the same for mine.

In your DDL.sql file, I noticed that some tables use ON DELETE CASCADE when they probably should not. For example, deleting a candle could also delete related sales history, which you may want to keep.

For everything else:

I did not notice any normalization issues.

I was able to import the SQL file without any problems. I also performed several queries to test the database, and everything worked correctly. All of the example data shown in the PDF is included in the INSERT statements. The SQL file is well structured and commented.

Your intersection table has the appropriate foreign keys and a primary key.

Other than the minor capitalization issue in the ERD, everything looked good. The PDF was easy to read, especially the overview, and it was well organized.

The feedback is entirely my own. I only used AI to improve clarity.

Alexander Nunez

Hello Zander & Abderrahmane,

Overall you have done a very good job on your outline and DDL file. I really enjoyed how you portrayed the shop keeper in the overview and gave us a very clear understanding of what this POS system is made to achieve.

Your schema does present a physical model that follows your outline and ER Diagram, but I do have to say that the ER Diagram is pretty hard to follow. There is consistency with

your outline and schema in terms of plural for entities and singular for attributes, but the ER diagram has the entities as singular still. Not a huge deal, but changing that would make it consistent across all diagrams and the outline. Capitalization is consistent among your entities and attributes as well. The schema itself is very clear and easy to read. It does a great job of providing a blueprint for your database. All intersection tables are formed correctly and include the necessary Foreign Keys. The schema seems to be in 3NF and the sample data provided does not indicate any kind of non-normalized issues or dependencies.

The SQL file seems to be syntactically correct and is formatted great! Having the SQL statements be all capitalized helps a lot with readability. All data types seem to be appropriate for the information they are meant to hold. All primary and foreign keys are correctly defined. I do believe ON DELETE CASCADE should be removed from the Sales table so as to not delete sales data when a customer becomes inactive or an employee leaves. All relationship tables are present and all sample data provided has been inserted. Overall, it's a great looking SQL file, but I would definitely make sure to comment more on it to help with clarity on some of your decisions or explain the sample data.

All feedback is my own.

Feifan Qi

Hi Group 54, I think your Project Step 2 draft is well organized and the database design fits the Candle Shop POS System well. The overview clearly explains the purpose of the system, and the selected entities represent the main parts of the business process.

Your schema appears to match the ER design. I like that you used the SalesItems intersection table to support the many-to-many relationship between Sales and Candles. Separating Scents and Sizes into their own entities is also a good choice because it reduces repeated data.

A few things I noticed while reviewing your draft:

The ER diagram is understandable, but some relationship lines are close together. Moving

some entities farther apart could make the diagram easier to read.

Your naming is mostly consistent. I would recommend checking that entity and attribute names match exactly across the overview, ER diagram, schema, and SQL files, including capitalization.

The SQL data types appear appropriate, and the primary keys and foreign keys seem to match the schema. The DDL file is also organized with comments, which makes it easier to follow.

One thing to review is the use of ON DELETE CASCADE for some foreign keys. For a POS system, some records such as sales history may need to remain even if related records are removed, so it is worth checking whether each cascade operation matches the intended behavior.

The sample data appears reasonably normalized, and I did not notice obvious partial dependency or transitive dependency issues.

Overall, your project is in good shape. My main suggestions are improving ER diagram readability and reviewing cascade delete choices.

This review was completed on my own without the use of AI.

David Jantz

Nice work! For simplicity, I just answered the questions in the order they were given on Canvas:

The schema presents a physical model that exactly follows the database outline and the ER logical diagram exactly.

There is consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming.

The schema is easy to read. The ERD is more tangled... If you have time, I would suggest straightening that out so there are few crossed lines.

Intersection tables are all properly formed.

Sample data looks good, does not suggest any non-normalized issues like partial dependencies or transitive dependencies.

The SQL file is syntactically correct -- I ran it locally on MySQL and all the queries ran smoothly to generate the database on my laptop.

In the SQL, data types are appropriate considering the description of the attribute in the database outline.

In the SQL, the primary and foreign keys are correctly defined when compared to the Schema. Appropriate CASCADE operations are declared.

In the SQL, relationship tables are present when compared to the ERD/Schema.

In the SQL, all example data shown in the PDF is in fact inserted -- I checked every table from the database the script generated for me locally.

The SQL is all well-structured and commented and appears to be hand-authored.

This review is entirely my work, no AI was used.

Identified Feedback	Action Taken	Details / Outcome
Overuse of ON DELETE CASCADE in SQL	Updated Constraints	Replaced CASCADE with ON DELETE RESTRICT for foreign keys in Candles, Sales, and SalesItems (candleId constraint). This enforces NOT NULL transaction rules while protecting historical sales revenue and inventory data from accidental deletion.
SQL Documentation & Comments	Added Detailed Comments	Added inline and block comments to DDL.sql for key table choices, including

		operational design decisions for Employees, Customers, Candles, Sales, and SalesItems.
ER Diagram (ERD) Line Untangling	Realigned & Uploaded New ERD	Spaced out entities in the visual diagram, eliminating crossed and tangled relationship lines to improve readability.
Singular vs. Plural Naming Consistency	Kept Intentional Design	Retained singular entity names for the logical ERD (Employee, Customer, etc.) and plural table names for the physical schema (Employees, Customers, etc.) as standard modeling practice.

		Updated narrative overview text to ensure logical references align with singular entities.
Capitalization & Formatting Mismatches	Standardized Overview Text	Cleaned up broken text lines in the PDF outline and aligned foreign key schema references to camelCase across all documentation.

Candle Shop Point Of Sale System

Team members: Zander Pantelick and Abderrahmane Rhandouri (Gandor)

Group 54

Overview:

The system we are designing is a POS system for a small Candle shop. The shop has moderate sales and regular customers in a small town. The store owner is a retired professional, and is pursuing his passion to produce and sell candles. His current volume is around 50 sales a day, with almost 10,000 customers as the whole town quenches its needs from him. However, despite the small number of sales, it is still lots for the senior to manage as he steps away from his high stress career for a more peaceful retirement. Through all the chaos of multiple spread sheets containing all information about sales and 30+ items in inventory, as well as his aging memory, where it keeps failing him with his customers' interactions. He decided that it's time to invest in a small budget POS system that will take care of all the headache that comes with managing the store numbers and data. As he relaxes to enjoy his retirement and his day to day friendly interactions with his customers. He decided that he needs a way to remember all details about customers, like their names and last names to make the interaction warmer. As well as a way to keep track of the number of candles he has and what type (size, scent etc...), the number of sales he made, and a way to register employees' information as he steps out sometimes, and to manage the hires he employs when it gets busy.

Data outline:

Employees:

This entity stores information about the employees in the shop so that we know the employee responsible for any sale, enabling issue resolution as well as performance tracking.

Relationships:

- **1:M**: One employee can process many sales, thus their ID will be referenced across multiple sales. An individual sale can only reference one employee.
- **(Implicit Many-to-Many)**: One employee can sell many candles, as they process sales containing multiple sales item lines, each referencing one candle entity.
- **(Implicit Many-to-Many)**: One employee can assist many different customers through shared sales records over time.

Attributes:

- **employeeId: int, PK, not NULL, AUTO_INCREMENT**
 - Unique ID. Stores the primary key, a unique identifier for the employee.
- **firstName: varchar(50), not NULL**
 - Stores the first name of the employee
- **lastName: varchar(50), not NULL**
 - Stores the last name of the employee
- **hireDate: date, NULL** stores the date when employee was hired
- **hireDate: date, NULL**
 - The date the employee was hired.

Customers:

This entity stores information about the customers in the shop. So that we know our customers, and track their information to know what they like and dislike, how much they have bought and their purchase history. As well as their contact information for future targeting or promotions and names and last names to have warmer interactions

Relationships:

- **1:M (One-to-Many)**: One customer can be part of many sales over their lifetime, meaning their unique ID will be referenced in multiple sales. However, an individual sale is tied explicitly to only one customer.
- **(Implicit Many-to-Many)**: One customer can purchase many different candles. This occurs because their customer ID is linked to sales, which consist of one or more sale items tracking different physical candles.
- **(Implicit Many-to-Many)**: One customer can interact with many different employees across multiple store visits, as derived via the shared sales records.

Attributes:

- **customerId: int, PK, not NULL, AUTO_INCREMENT**
 - Unique ID. Stores the primary key, a unique identifier for the customer.
- **firstName: varchar(50), not NULL**
 - Stores the first name of the customer.
- **lastName: varchar(50), not NULL**
 - Stores the last name of the customer.
- **email: varchar(100), UNIQUE INDEX, NULL**
 - Unique email constraint. Stores the customer's email address and strictly prevents duplicate accounts from being registered under the exact same email.
- **phone: varchar(15), NULL**
 - Stores the customer's telephone number for contact purposes.
- **createdDate: date, not NULL, DEFAULT (CURRENT_DATE)**
 - Stores the registration date of the customer profile.
- **isActive: tinyint(1), not NULL, DEFAULT 1**
 - Flag used for soft-deleting customer records without removing historical sales logs.

Candles:

This entity serves as the core physical product catalog. It keeps track of the specific inventory on hand and establishes the exact product configurations available for sale by joining scent and size variations.

Relationships:

- **M:1 (Many-to-One):** Many distinct candles can share the same scent profile (e.g., a "Vanilla" scent available in small, medium, or large shapes), but a single candle SKU must map to exactly one scent.
- **M:1 (Many-to-One):** Many distinct candles can share the same size classification (e.g., multiple different scents poured into 8 oz jars), but a single candle SKU must map to exactly one size.
- **1:M (One-to-Many):** One candle can be added to many different sale item lines across different customer receipts. A single line item in a receipt, however, isolates exactly one candle SKU.

Attributes:

- **candleId: int, PK, not NULL, AUTO_INCREMENT**
 - Unique ID. Stores the primary key, a unique stock-keeping identifier (SKU) for the finished candle product.
- **scentId: int, FK, not NULL**
 - Foreign Key reference. Links directly to Scents(scentId) to specify the candle's scent category. Cannot be null.

- **sizeId: int, FK, not NULL**
 - Foreign Key reference. Links directly to Sizes(sizeId) to specify the candle's structural size category. Cannot be null.
- **price: decimal(6,2), not NULL**
 - Standard base price format. Stores the baseline retail cost of the product before transaction-specific discounts or taxes.
- **stockQty: int, not NULL**
 - Inventory count. Stores the current volume of physical items actively sitting on store shelves or in storage.
- **lastUpdated: timestamp, not NULL, DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP**
 - Tracks last time the entity was updated.

Sales:

This entity acts as the transaction header log. It registers the higher-level financial exchange event, tracking exactly when an exchange happened, who processed it, who bought it, and the total monetary amount received.

Relationships:

- **M:1 (Many-to-One):** Many sales can be handled by a single employee, but a single sales record can explicitly reference only one responsible employee.
- **M:1 (Many-to-One):** Many sales can belong to a single customer over time, but an individual sales receipt maps to exactly one customer.
- **1:M (One-to-Many):** One sales transaction or record must contain at least one or more line items (SalesItems) representing the products purchased. A single line item points back to exactly one parent sales invoice.

Attributes:

- **saleId: int, PK, not NULL, AUTO_INCREMENT**
 - Unique ID. Stores the primary key, a unique receipt or invoice tracking number for the transaction event.
- **saleTimestamp: timestamp, not NULL, DEFAULT CURRENT_TIMESTAMP**
 - Event timestamp. Stores the exact chronological date and time when the sale was completed.
- **totalAmount: decimal(8,2), not NULL**
 - Financial total. Stores the exact final dollar amount paid at register checkout, including calculated sales tax and active promotions.
- **employeeId: int, FK, not NULL**
 - Foreign Key reference. Links directly to Employees(employeeId) to log the staff member handling checkout. Cannot be null.
- **customerId: int, FK, not NULL**
 - Foreign Key reference. Links directly to Customers(customerId) to log the purchasing customer. Cannot be null.

Scents:

This entity manages the shop's catalog of scent profiles. It decouples descriptive sensory information from physical inventory management, ensuring uniform naming conventions across your store website and backend tools.

Relationships:

- **1:M (One-to-Many):** One scent profile can be utilized across many different candles of varying physical sizes. A candle product config, however, is bound to exactly one scent category.

Attributes:

- **scentId: int, PK, not NULL, AUTO_INCREMENT**
 - Unique ID. Stores the primary key, a unique classification number for the olfactory profile.
- **scentName: varchar(100), not NULL**
 - Scent name designation. Stores the commercial name of the scent profile (e.g., 'Lavender Fields', 'Teakwood & Amber').
- **description: varchar(255), NULL**
 - Descriptive notes. Stores customer-facing or internal metadata detailing the fragrance notes and scent characteristics.

Sizes:

This entity acts as a category master table governing the product sizing specs. It centralizes uniform dimensions, volumetric data, and label names to prevent data entry spelling anomalies.

Relationships:

- **1:M (One-to-Many):** One size category can apply to many different individual candles across various scent lines. A specific candle is built to fit exactly one size profile.

Attributes:

- **sizeId: int, PK, not NULL, AUTO_INCREMENT**
 - Unique ID. Stores the primary key, a unique numerical code for the size profile.
- **sizeLabel: varchar(20), not NULL**
 - Market label. Stores the retail name of the size (e.g., 'Small Tin', 'Large 3-Wick').
- **volumeOz: decimal(4,1), not NULL**
 - Physical capacity. Stores the precise liquid or weight mass calculation in

fluid ounces to power catalog filters.

SalesItems:

This is the transaction line item bridge entity. Because a single sale invoice can contain multiple items, and a single type of candle is sold across many separate customer visits, this table maps the Many-to-Many (M:N) relationship between Sales and Candles, capturing itemized transaction specifics.

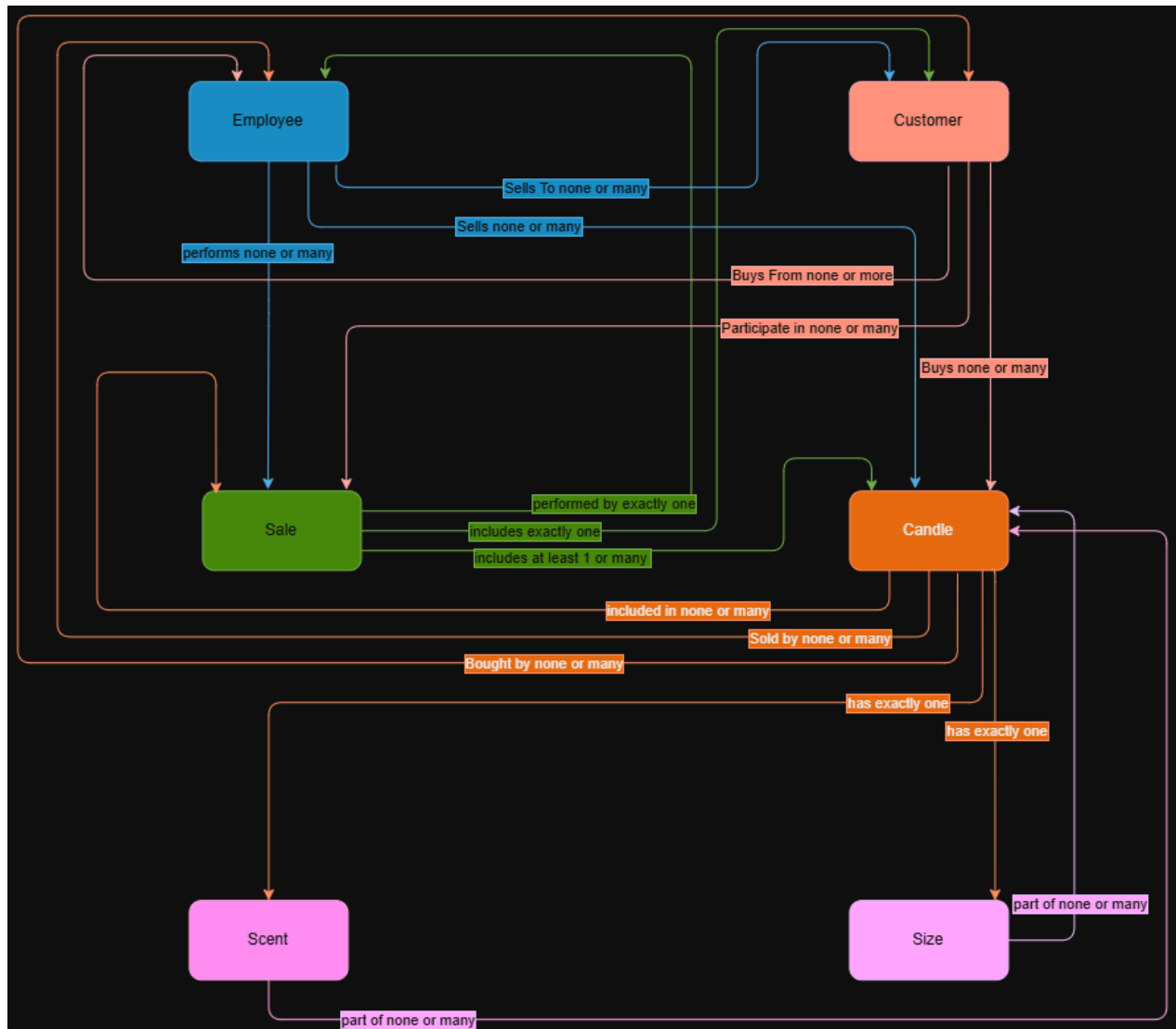
Relationships:

- **M:1 (Many-to-One):** Many individual line items can roll up into one parent sale receipt. However, an individual line item belongs to exactly one transaction.
- **M:1 (Many-to-One):** Many line items across different store receipts can reference the exact same candle model, but a single row in this table isolates exactly one unique candle ID.

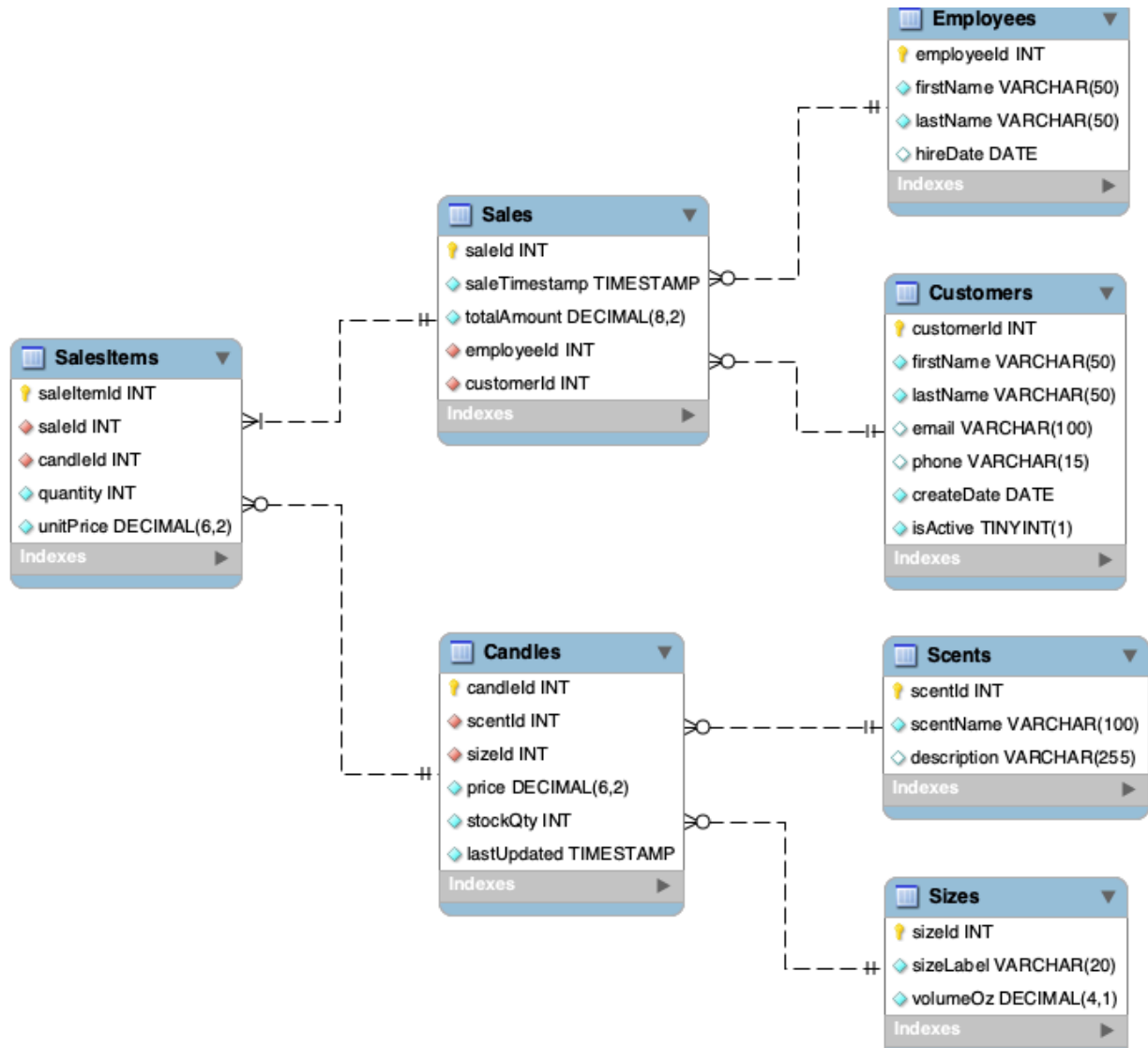
Attributes:

- **saleItemId: int, PK, not NULL, AUTO_INCREMENT**
 - Unique Line ID. Stores the primary key, a unique identifier separating this specific item line from others.
- **saleId: int, FK, not NULL**
 - Foreign Key reference. Links directly to Sale(saleId) to associate the line item with its overarching invoice receipt. Cannot be null.
- **candleId: int, FK, not NULL**
 - Foreign Key reference. Links directly to Candles(candleId) to specify exactly which candle product was bought on this line. Cannot be null.
- **quantity: int, not NULL**
 - Order volume. Stores the total count of this specific candle model purchased inside this specific transaction.
- **unitPrice: decimal(6,2), not NULL**
 - Locked historical price. Stores the exact cost per unit at the literal moment of purchase, preserving historical accounting records if base prices change in the Candles table later.

Entity Relationship Diagram:



Schema:



Sample Data:

employeeId	firstName	lastName	hireDate
1	John	Doe	2024-01-15
2	Jane	Smith	2024-03-01
3	Robert	Johnson	2025-06-10

customerId	firstName	lastName	email	phone	createdAt	isActive
1	Alice	Walker	alice.walker@example.com	541-555-0192	2025-01-10	1
2	Charlie	Brown	charlie.b@example.com	541-555-0144	2025-03-22	1
3	Emma	Davis	emma.davis@example.com	541-555-0188	2025-11-05	0

scentId	scentName	description
1	Lavender Fields	Soothing fresh lavender with subtle notes of eucalyptus.

2	Teakwood & Amber	Warm woody aroma blended with sweet amber and spice.
3	Vanilla Bean	Classic creamy vanilla with a soft, sweet finish.

sizeId	sizeLabel	volumeOz
1	Small Tin	4.0
2	Medium Jar	8.0
3	Large 3-Wick	16.0

candleId	scentId	sizeId	price	stockQty	lastUpdated
1	1	1	12.50	25	2026-07-23 12:00:00
2	1	2	22.00	15	2026-07-23 12:00:00
3	2	2	24.00	10	2026-07-23 12:00:00
4	3	3	35.00	8	2026-07-23 12:00:00

saleId	saleTimestamp	totalAmount	employeeId	customerId
1	2026-07-20 10:15:00	34.50	1	1
2	2026-07-20 11:30:00	35.00	1	2
3	2026-07-21 14:05:00	44.00	2	1

saleItemId	saleId	categoryId	quantity	unitPrice
1	1	1	1	12.50
2	1	2	1	22.00
3	2	4	1	35.00
4	3	2	2	22.00

Citations:

Used Gemini AI for Sample data to populate our tables.

Step 3: Gemini was used to fill in filler words/ descriptions and sample data for front end ui.